Probability toolkit: **Total prob:** $P(A) = \sum_k P(A|B_k)P(B_k)$. **Bayes:** $P(B_k|A) = P(A|B_k)P(B_k)/\sum_j P(A|B_j)P(B_j)$. **Joint:** $p(x,z) = p(x|z)p(z)$. **Marginal:** $p(x) = \int p(x,z)\,dz$ (or $\sum_z$). **Conditional indep.** given θ: $p(\tilde{x}|\theta,x) = p(\tilde{x}|\theta)$. **Indicator:** $\mathbb{E}[I(A)] = P(A)$.

Tower: $\mathbb{E}X = \mathbb{E}[\mathbb{E}(X|Z)]$. **Total variance:** $\mathrm{Var}X = \mathbb{E}[\mathrm{Var}(X|Z)] + \mathrm{Var}[\mathbb{E}(X|Z)]$. **Linear:** $\mathrm{Var}(aX+bY) = a^2\mathrm{Var}X + b^2\mathrm{Var}Y + 2ab\mathrm{Cov}(X,Y)$; $\mathrm{Var}(t^\top\hat{\theta}) = t^\top V t$; $\mathrm{Var}(AY) = A\mathrm{Var}(Y)A^\top$.

Square trick (CAVI): $\mathbb{E}[(c-\mu)^2] = (c-\mathbb{E}\mu)^2 + \mathrm{Var}\mu$, so $\mathbb{E}[\mu^2] = (\mathbb{E}\mu)^2 + \mathrm{Var}\mu$.

Jensen: f convex $\Rightarrow \mathbb{E}f(X) \geq f(\mathbb{E}X)$. Log is concave: $\log \mathbb{E}Y \geq \mathbb{E}\log Y$, equal iff Y is constant.

Invariance: MLE of $g(\theta)$ is $g(\hat{\theta})$. **Delta:** $g(\hat{\theta}) \approx \mathrm{N}(g(\theta), g'(\theta)^2 I(\theta)^{-1})$. **Taylor:** $g(Y) \approx g(\mu) + (Y-\mu)g'(\mu)$.

Integrals: $\int_0^\infty x^{a-1}e^{-bx}dx = \Gamma(a)/b^a$; $\int_0^1 x^{a-1}(1-x)^{b-1}dx = B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$; $\Gamma(a+1) = a\Gamma(a)$; $\Gamma(n) = (n-1)!$; $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

Inverse: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. **Odds:** $o = \frac{p}{1-p}$, $p = \frac{o}{1+o}$.

Complete the square: $-\frac{A}{2}\mu^2 + B\mu$ is the kernel of $\mathrm{N}(B/A,\ 1/A)$. Course log = ln: press **ln** on the FX-82, not log. Round final answers to 2dp.

Distributions NOT on formula sheet:

Name	pmf / pdf	Mean; Variance
Bin(m,p)	$\binom{m}{x}p^x(1-p)^{m-x}$	mp; $mp(1-p)$
Geo $x\geq 1$	$p(1-p)^{x-1}$	$1/p$; $(1-p)/p^2$
Geo $x\geq 0$	$p(1-p)^x$	$(1-p)/p$; $(1-p)/p^2$
NB(r,p)	$\binom{x+r-1}{x}p^r(1-p)^x$	$\frac{r(1-p)}{p}$; $\frac{r(1-p)}{p^2}$
Beta(a,b)	$\frac{x^{a-1}(1-x)^{b-1}}{B(a,b)}$	$\frac{a}{a+b}$; $\frac{ab}{(a+b)^2(a+b+1)}$

Variances the sheet omits: Exp(rate λ) $1/\lambda^2$; Gamma$(a,$rate $b)$ a/b^2; InvGamma(a,b) $\frac{b^2}{(a-1)^2(a-2)}$ $(a>2)$; Poisson λ.

Gamma extras: $\mathbb{E}[1/X] = b/(a-1)$ $(a>1)$; $\mathbb{E}[\log X] = \psi(a) - \log b$, $\psi = \Gamma'/\Gamma$. Scale form: $x^{a-1}e^{-x/b}/(\Gamma(a)b^a)$ has mean ab. $\tau \sim \mathrm{Gamma}(a,b) \iff 1/\tau \sim \mathrm{InvGamma}(a,b)$; if $\sigma^2 \sim \mathrm{InvGamma}(a,b)$, $\mathbb{E}[1/\sigma^2] = a/b$.

Bivariate Normal: $f(x) = \frac{1}{2\pi|\Sigma|^{1/2}}\exp\{-\frac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu)\}$. $X_1|X_2{=}x_2 \sim \mathrm{N}(\mu_1 + (x_2-\mu_2)\sigma_{12}/\sigma_2^2,\ \sigma_1^2 - \sigma_{12}^2/\sigma_2^2)$.

Facts: $Z^2 \sim \chi_1^2$; sum of k indep. $Z^2 \sim \chi_k^2$ (mean k, var $2k$); $Z/\sqrt{U/r} \sim t_r$; $(U/m)/(V/n) \sim F_{m,n}$; $t_\nu^2 \sim F_{1,\nu}$. $\sum \mathrm{Pois}(\lambda_i)$ is $\mathrm{Pois}(\sum\lambda_i)$; $\sum_1^n \mathrm{Exp}(\lambda)$ is $\mathrm{Gamma}(n,\lambda)$.

Critical values: z: 1.645 (90%), **1.96** (95%), 2.576 (99%). $\chi^2_{df}(.95)$: df 1: **3.84**, 2: 5.99, 3: 7.81, 4: 9.49, 5: 11.07, 6: 12.59, 8: 15.51, 10: 18.31. $\chi^2_{df}(.99)$: 1: 6.63, 2: 9.21, 3: 11.34. $\Phi(1) = .84$, $\Phi(2) = .98$. $t_{60}(.975) = 2.00$.

Confidence intervals: estimate $\pm$ critical value $\times$ SE. **Pivot:** function of data and θ with fully known distribution.

Want	95% CI
μ, σ known	$\bar{x} \pm z\sigma/\sqrt{n}$
μ, σ unk.	$\bar{x} \pm t_{n-1}\,s/\sqrt{n}$
$\mu_X - \mu_Y$, equal var	$\bar{x}-\bar{y} \pm t_{n+m-2}\,s_p\sqrt{1/n+1/m}$
p	$\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$
σ^2 (χ^2_{n-1} c)	$[(n-1)s^2/c_{.975},\ (n-1)s^2/c_{.025}]$
New X^* (PI)	$\bar{x} \pm t_{n-1}\,s\sqrt{1+1/n}$

Say: "If we repeated this, 95% of such intervals would contain the true value." Do not say "95% chance θ is in this interval" (that is a Bayesian credible interval). CI excludes 0 (or 1 for a ratio) $\iff$ reject H_0 at 5%.

Hypothesis tests: H_0: no effect. H_1: effect. **Test statistic:** known distribution under H_0. Reject if statistic beyond critical value, or $p < \alpha$. **p-value:** P(result at least this extreme | H_0 true), not P(H_0 true). **Type I:** reject true H_0 (prob α). **Type II:** keep false H_0 (prob β). **Power:** $1-\beta$. Write "reject" or "insufficient evidence to reject", never "accept". χ^2 and F tests are right-tailed. Statistical $\neq$ practical significance.

Estimators: **Bias** $= \mathbb{E}\hat{\theta} - \theta$. **MSE** $= \mathrm{Var}\hat{\theta} + \mathrm{Bias}^2$. **Unbiased:** Bias 0. **Consistent:** $\hat{\theta} \to \theta$ as $n \to \infty$. **Standard error:** estimated sd of $\hat{\theta}$. **Precision:** 1/variance. $\bar{X} \sim \mathrm{N}(\mu, \sigma^2/n)$, and approximately so for large n by the CLT. $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ for Normal data.

Model terms: **Nested:** A comes from B by constraining parameters (usually to 0). **Factor** with L levels: $L-1$ dummies vs reference level; test it by one LRT on $L-1$ df, not its separate z values. `x1*x2` = `x1+x2+x1:x2`. `I(x^2)`: quadratic term. Coefficients mean "holding other predictors fixed". Observational data show **association, not causation**. A predictor can flip sign when correlated with others: it is "not significant in the presence of the others".

Likelihood: independent Y_i with densities $f_i(\cdot;\theta)$ (may depend on covariates): $l(\theta) = \sum_i \log f_i(y_i;\theta)$. MLE $\hat{\theta}$ maximises l.

Score: $U(\theta) = \partial l/\partial\theta$. **Observed information:** $J(\theta) = -\partial^2 l/\partial\theta\,\partial\theta^\top = -H$, depends on y. **Fisher information:** $I(\theta) = \mathbb{E}J(\theta;Y)$, no y. This is the whole sample ($n\times$ one-obs info if iid). Bigger information = sharper curvature = smaller variance.

Asymptotics ($n\to\infty$, θ^* true): **Consistent:** $P(|\hat{\theta} - \theta^*| > \epsilon) \to 0$. **Normal:** $\hat{\theta} \approx \mathrm{N}(\theta^*, I(\theta^*)^{-1})$. **Efficient:** asymptotically unbiased with smallest variance $I(\theta^*)^{-1}$.

Regularity: l smooth (3rd derivatives exist, continuous, bounded expectation); support of Y free of θ; parameter space finite-dimensional, free of Y; θ^* not on boundary.

Q1 recipe: (a) l. (b) l', l''; $I = -\mathbb{E}[l'']$, replace $\sum x_i$ by $n\mathbb{E}X$ if data remain. (c) Solve $l' = 0$, check $l'' < 0$, state $\hat{\theta} \approx \mathrm{N}(\theta, 1/I(\theta))$.

iid $X_i \sim$	MLE	$I(\theta)$
Exp(rate λ)	$1/\bar{x}$	n/λ^2
Exp(mean λ)	$\bar{x}$	n/λ^2
Poisson(λ)	$\bar{x}$	n/λ
N$(\mu_0, 1/\tau)$	$n/\sum(x_i-\mu_0)^2$	$n/(2\tau^2)$
N(μ_0, σ^2)	$\sum(x_i-\mu_0)^2/n$	$n/(2\sigma^4)$
N(μ, σ_0^2)	$\bar{x}$	n/σ_0^2
Bin(m,p)	$\bar{x}/m$	$\frac{nm}{p(1-p)}$
Geom $x\geq 1$	$1/\bar{x}$	$\frac{n}{p^2(1-p)}$
Gamma$(a$ kn.$,\lambda)$	$a/\bar{x}$	na/λ^2
$\kappa(1+x)^{-\kappa-1}$	$n/\sum\log(1+x_i)$	n/κ^2
NegBin$(r,1-p)$, one y	$y/(y+r)$	$\frac{r}{p(1-p)^2}$

Wald CI for $t^\top\theta$: $\hat{\theta} \approx \mathrm{N}(\theta^*, I(\hat{\theta})^{-1})$, so $t^\top\hat{\theta} \pm z_\alpha\sqrt{t^\top I(\hat{\theta})^{-1}t}$, $\Phi(z_\alpha) = 1-\alpha/2$. Unit vector t: $\hat{\theta}_i \pm z_\alpha\sqrt{[I(\hat{\theta})^{-1}]_{ii}}$. No I? Use $J(\hat{\theta};y)$.

LR confidence region: $\{\theta : 2l(\hat{\theta}) - 2l(\theta) \leq \chi^2_k(1-\alpha)\}$, $k = \dim\theta$. Better than the Normal approximation for small n.

Wald test: $H_0: \beta_j = 0$. $z^* = \hat{\beta}_j/\mathrm{se}(\hat{\beta}_j) \sim \mathrm{N}(0,1)$ under H_0. Reject if $|z^*| > 1.96$.

LRT: Reduced R nested in Full F, k fewer parameters. $LR^* = -2[\log L(\hat{\theta}^R) - \log L(\hat{\theta}^F)] \sim \chi^2_k$ under R. If $LR^* > \chi^2_k(1-\alpha)$ reject R, pick F.

Wald vs LRT: asymptotically equivalent ($z^{*2} \approx \chi^2_1$), but the χ^2 approximation to the LR is generally better, so prefer LRT.

Binomial regression: $Y_i \sim \mathrm{bin}(m_i, p_i)$ independent, m_i known (binary: $m_i = 1$), $g(p_i) = \eta_i = x_i^\top\beta$. Linear $p = \beta_0 + \beta_1 t$ fails because $0 \leq p \leq 1$ is needed.

Logit: $\eta = \log\frac{p}{1-p}$, $p = \frac{e^\eta}{1+e^\eta} = \frac{1}{1+e^{-\eta}}$. **Probit:** $\eta = \Phi^{-1}(p)$, $p = \Phi(\eta)$. **Cloglog:** $\eta = \log(-\log(1-p))$, $p = 1-e^{-e^\eta}$.

Logistic curve $\eta = \beta_0 + \beta_1 t$: $p = \frac{1}{2}$ at $t = -\beta_0/\beta_1$ (location), slope there $\beta_1/4$ (steepness).

$$l(\beta) = c + \sum_i \left[y_i \log\tfrac{p_i}{1-p_i} + m_i\log(1-p_i)\right]$$

Logit: $l = c + \sum_i[y_i\eta_i - m_i\log(1+e^{\eta_i})]$. No closed form: use `optim` or IWLS.

Logit information: $J = I = X^\top WX$, $W = \mathrm{diag}(m_i p_i(1-p_i))$ (no y_i left). One predictor:

$$I(\beta) = \begin{pmatrix} \sum m_i p_i q_i & \sum m_i x_i p_i q_i \\ \sum m_i x_i p_i q_i & \sum m_i x_i^2 p_i q_i \end{pmatrix},\ q_i = 1-p_i$$

Probit: $I = X^\top WX$, $W_{ii} = m_i\varphi(\eta_i)^2/[\Phi(\eta_i)(1-\Phi(\eta_i))]$.

Predict at x_0: $\hat{\eta}_0 = x_0^\top\hat{\beta}$, $\hat{p}_0 = g^{-1}(\hat{\eta}_0)$.

CI for p: p is not linear in β, so (1) CI for η: $\hat{\eta}_0 \pm 1.96\sqrt{x_0^\top V x_0}$, $V = I(\hat{\beta})^{-1}$; one predictor $x_0^\top V x_0 = V_{11} + 2x_0V_{12} + x_0^2V_{22}$. (2) Apply g^{-1} to both ends (g monotone).

Odds: $\log o = \eta$. Change x_j by Δ: $OR = e^{\Delta\beta_j}$, CI $\exp\{\Delta(\hat{\beta}_j \pm 1.96\,\mathrm{se})\}$ (swap ends if $\Delta < 0$). Several x change by vector d: $\log OR = d^\top\hat{\beta}$, se $= \sqrt{d^\top V d}$.

Classify: $\hat{p} \geq$ cutoff; trade off false positives and negatives by cost (screening: fewer false negatives).

Deviance: **Saturated S:** one parameter per observation, $\hat{\mu}_i = y_i$. **Null:** intercept only. **Scaled deviance:** $D^A/\phi = -2\log[L(\hat{\beta}_A)/L(S)] = 2(l_S - l_A)$. **Deviance** is $D^A = \sum d_i$. Binomial and Poisson: $\phi = 1$, same thing. $D_{null} - D_{res}$ = improvement from predictors.

Why (Poisson): $l_S = \sum[y_i\log y_i - y_i - \log y_i!]$, $l = \sum[y_i\log\hat{\mu}_i - \hat{\mu}_i - \log y_i!]$; double the gap.

Model	d_i (0 log 0 = 0)
Normal	$(y_i-\hat{\mu}_i)^2$ (= SSE in LM)
Poisson	$2[y_i\log(y_i/\hat{\mu}_i) - (y_i-\hat{\mu}_i)]$
Binomial	$2[y_i\log\frac{y_i}{\hat{\mu}_i} + (m_i-y_i)\log\frac{m_i-y_i}{m_i-\hat{\mu}_i}]$
Gamma	$2[-\log(y_i/\hat{\mu}_i) + (y_i-\hat{\mu}_i)/\hat{\mu}_i]$

Adequacy (goodness of fit), $\phi = 1$ only: H_0: fitted model adequate. If $m_ip_i,\ m_i(1-p_i) \geq 5$ (binomial) or $\lambda_i \geq 5$ (Poisson) then $D \approx \chi^2_{n-p}$, p including β_0. Reject if $D > \chi^2_{n-p}(.95)$; rough check $D \approx df$. Saturated model has n parameters (not fixed), so ML theory does not apply: D need not be χ^2. Small m_i or counts (binary $m_i = 1$ especially): no adequacy test, ΔD still fine. Normal, Gamma, IG: ϕ estimated, cannot test adequacy.

Lack of fit may be overdispersion, wrong mean or missing interaction: check residuals vs $\hat{\eta}$ first.

Nested: A in B, s extra parameters. $(D^A - D^B)/\phi = -2[l_A - l_B]$, l_S cancels. $\phi = 1$: $D^A - D^B \approx \chi^2_s$. ϕ unknown: F test (see Overdispersion). Subtract residual deviances, not null deviances.

AIC (non-nested too): $AIC = 2p - 2\hat{l}$, smaller is better, same data only, not a test. B has s more parameters: $AIC_B - AIC_A = 2s - (D_A - D_B)/\phi$ (exact for $\phi = 1$).

From R output: $p = n-$residual df; $n =$ null df $+1$. $\phi = 1$: $\hat{l} = p - AIC/2$, $l_S = \hat{l} + D/2$ (same l_S for every model on those data). Gamma, Normal: R's AIC counts ϕ, so $\hat{l} = (p+1) - AIC/2$.

Exponential family: $f(y;\theta,\phi) = \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right]$. θ canonical (location), ϕ dispersion (scale). Form not unique. Convention $a(\phi) = \phi$.

$\mathbb{E}Y = b'(\theta)$, $\mathrm{Var}Y = b''(\theta)a(\phi) = v(\mu)a(\phi)$. **Variance function** $v(\mu) = b''((b')^{-1}(\mu))$: how spread changes with mean. **Canonical link** $g = (b')^{-1}$, $g(\mu) = \theta$.

Proof: $\int f\,dy = 1$. Differentiate in θ: $\mathbb{E}[(Y-b')/a] = 0$, so $\mathbb{E}Y = b'$. Again: $\mathbb{E}[(Y-b')^2/a^2] - b''/a = 0$, so $\mathrm{Var}Y = b''a$.

Recipe: write $\log f$; find $y\times$(something) $= \theta/a$; the term with no y gives $-b/a$; leftovers c. Write b in θ only. Check $b'(\theta) = \mu$. State all of θ,ϕ,a,b,c. Not exp family if the support depends on θ, or the parameter multiplies y^2, $\log y$ or e^y rather than y.

- **Normal:** $\theta = \mu$, $\phi = \sigma^2$, $b = \theta^2/2$, $c = -\frac{1}{2}(\frac{y^2}{\phi} + \log 2\pi\phi)$; $g = \mu$, $v = 1$.
- **Poisson:** $\theta = \log\lambda$, $\phi = 1$, $b = e^\theta$, $c = -\log y!$; $g = \log\mu$, $v = \mu$.
- **Binomial:** $\theta = \log\frac{p}{1-p}$, $\phi = 1$, $b = m\log(1+e^\theta)$, $c = \log\binom{m}{y}$; $g = \log\frac{\mu}{m-\mu}$, $v = \mu(1-\mu/m)$.
- **Gamma**(ν,λ): $\theta = -\lambda/\nu = -1/\mu$, $\phi = 1/\nu$, $b = -\log(-\theta)$, $c = (\frac{1}{\phi}-1)\log y - \frac{\log\phi}{\phi} - \log\Gamma(\frac{1}{\phi})$; $g = -1/\mu$ (R: $1/\mu$), $v = \mu^2$, $\mathrm{Var}Y = \phi\mu^2$.
- **Exponential**(λ): $\theta = -\lambda$, $\phi = 1$, $b = -\log(-\theta)$, $c = 0$; $g = -1/\mu$, $v = \mu^2$. $\mu = -1/\eta$ needs $\eta < 0$, so canonical link can fail: use log. ϕ known: χ^2 tests.
- **Inverse Gaussian:** $\theta = -\frac{1}{2\mu^2}$, $\phi = 1/\lambda$, $b = -\sqrt{-2\theta}$, $c = -\frac{1}{2}[\log(2\pi\phi y^3) + \frac{1}{\phi y}]$; $g = -\frac{1}{2\mu^2}$ (R: $1/\mu^2$), $v = \mu^3$.
- **Weibull,** k known: $Y = X^k \sim \mathrm{Exp}(\mathrm{rate}\ \lambda^k)$.

GLM: Y from exponential family and $\mu = \mathbb{E}Y = g^{-1}(x^\top\beta)$; g monotone differentiable link. We model location $\eta = x^\top\beta$; scale sorts itself out. Canonical link: $\theta_i = x_i^\top\beta$.

Fitted mean: log link e^η; logit $me^\eta/(1+e^\eta)$; identity η; inverse $1/\eta$.

Newton-Raphson: $x_{n+1} = x_n - f'(x_n)/f''(x_n)$; $\theta^{n+1} = \theta^n - H(\theta^n)^{-1}U(\theta^n)$. **Fisher scoring:** replace J by I (positive definite): $\theta^{n+1} = \theta^n + I(\theta^n)^{-1}U(\theta^n)$. Equal under a canonical link.

Score: $U_j = \sum_i \frac{(y_i-\mu_i)x_{ij}}{a(\phi)v(\mu_i)g'(\mu_i)}$. Canonical link: $X^\top(y-\hat{\mu}) = 0$, so $\sum\hat{\mu}_i = \sum y_i$ with an intercept.

WLS: $Y = X\beta + \varepsilon$, $\varepsilon \sim \mathrm{N}(0,\Sigma)$: $\hat{\beta} = (X^\top\Sigma^{-1}X)^{-1}X^\top\Sigma^{-1}y$.

Idea: $g(Y_i) \approx Z_i = g(\mu_i) + (Y_i-\mu_i)g'(\mu_i) = x_i^\top\beta + \varepsilon_i$, $\mathrm{Var}\varepsilon_i = g'(\mu_i)^2 v(\mu_i)a(\phi)$, diagonal Σ. Depends on β, so iterate.

IWLS: (1) $\hat{\mu}^0 = y$. (2) $z_i = g(\hat{\mu}_i) + (y_i - \hat{\mu}_i)g'(\hat{\mu}_i)$, $W_{ii} = 1/[g'(\hat{\mu}_i)^2 v(\hat{\mu}_i)]$. (3) $\hat{\beta} = (X^\top WX)^{-1}X^\top Wz$, $\hat{\mu}_i = g^{-1}(x_i^\top\hat{\beta})$. (4) Repeat until $\hat{\beta}$ stable. $a(\phi)$ cancels. Same as Fisher scoring. Needs only g, v.

Family, link	z_i	W_{ii}
Poisson, log	$\eta + (y-\mu)/\mu$	μ
Binom, logit	$\eta + \frac{(y-\mu)m}{\mu(m-\mu)}$	$\mu(m-\mu)/m$
Gamma, log	$\eta + (y-\mu)/\mu$	1

Variance: $\mathrm{Var}\hat{\beta} = (X^\top\hat{\Sigma}^{-1}X)^{-1}$, $\hat{\Sigma}_{ii} = g'(\hat{\mu}_i)^2 v(\hat{\mu}_i)a(\hat{\phi})$. $a(\phi)$ does not cancel here.

Estimate ϕ: $\hat{\phi} = X^2/(n-p)$, Pearson $X^2 = \sum_i(y_i-\hat{\mu}_i)^2/v(\hat{\mu}_i)$. Consistent. Not $D/(n-p)$. If $\phi = 1$ and the model is adequate, $X^2 \approx \chi^2_{n-p}$.

Rate model: $Y_i \sim \mathrm{Pois}(\lambda_i)$, $\lambda_i \propto t_i$: $\log\lambda_i = \log t_i + x_i^\top\beta$. `offset(log(t))` fixes the coefficient of $\log t$ at 1 (uses no df). e^β = rate ratio; $\hat{\lambda} = te^{x^\top\hat{\beta}}$.

Gamma GLM: `Gamma(link="log")` etc. ϕ estimated, so t and F. sd/$\mu = \sqrt{\phi}$. Pick link by plotting $g(y)$ vs x.

Residuals: **Response:** $y_i - \hat{\mu}_i$; not homoskedastic unless v constant. **Pearson:** $r_P(i) = (y_i - \hat{\mu}_i)/\sqrt{v(\hat{\mu}_i)}$, $\sum r_P^2 = X^2$. **Deviance:** $r_D(i) = \mathrm{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}$, $\sum r_D^2 = D$ (R default).

Leverage: $H = X(X^\top\hat{\Sigma}^{-1}X)^{-1}X^\top\hat{\Sigma}^{-1}$, H_{ii} measures potential influence (unusual x; also depends on IWLS weight), no ϕ. $\sum H_{ii} = p$: rule of thumb, flag $> 2p/n$. Large leverage need not mean actual influence.

Studentised: $\mathrm{Var}(Z - \hat{Z}) = (I-H)\Sigma$, so $r_{SP} = r_P/\sqrt{(1-H_{ii})\hat{\phi}}$, $r_{SD} = r_D/\sqrt{(1-H_{ii})\hat{\phi}}$. Unit variance after adjusting for leverage. $\pm$: above/below fit.

Jack-knife: change in $\hat{\mu}_i$ when y_i removed (outliers): $\mathrm{sign}(y_i - \hat{\mu}_i)\sqrt{(1-H_{ii})r_{SD}^2 + H_{ii}r_{SP}^2}$.

Cook: change in the whole fit when i removed: $(\hat{\beta}_{(i)} - \hat{\beta})^\top X^\top\hat{\Sigma}^{-1}X(\hat{\beta}_{(i)} - \hat{\beta})/p \approx r_{SP}^2 H_{ii}/[p(1-H_{ii})]$.

Half-normal plot: ordered |values| vs $\Phi^{-1}(\frac{n+i}{2n+1})$. Smooth = fine; jump or kink in tail = problem. Need not be straight.

r_D **vs** $\hat{\eta}$: no trend, even spread is good. Trend or curve: wrong mean. Fan: wrong variance. **Banding:** discrete counts, not a problem; use a smoother.

Linearity: plot $g(y)$ or z vs each x_j: want straight. Fix by transforming x, not y (breaks distribution of Y).

Influential point: refit without it, compare $\hat{\beta}$, D, $\hat{\phi}$. Typically keep unless a recording error.

Overdispersion: actual $\mathrm{Var}Y_i$ larger than the model says. **Causes:** binomial: dependent trials or p changing trial to trial. Poisson: dependent events or non-constant rate.

Detect: $D \gg df$, GOF p tiny, $\hat{\phi} \gg 1$ (no universal cutoff). Rule out wrong mean and outliers first.

If ignored: $\hat{\beta}$ unaffected, but SEs too small, z too big, p-values too small, CIs too narrow: spurious significant predictors, too big a model.

Fixes: quasi-likelihood; different model (Poisson with Gamma rate gives NegBin, Var $= \mu + \mu^2/\nu$); Bayes: random effect in η.

Quasi-likelihood: specify only $g(\mu_i) = x_i^\top\beta$, $\mathbb{E}Y_i = \mu_i$, $\mathrm{Var}Y_i = \phi v(\mu_i)$; no distribution. Score $u_i = \frac{Y_i-\mu_i}{\phi v(\mu_i)}$: $\mathbb{E}u = 0$, $\mathrm{Var}u = \frac{1}{\phi v}$, $-\mathbb{E}\frac{\partial u}{\partial\mu} = \frac{1}{\phi v} = \mathbb{E}u^2$, same as $\partial\log L/\partial\mu$.

$$Q_i = \int_{y_i}^{\mu_i}\frac{y_i - t}{\phi v(t)}dt, \quad Q = \sum_i Q_i$$

$\partial Q_i/\partial\mu_i = u_i$. Q need not be a real log-density. $\hat{\beta}$ is consistent and asymptotically Normal as for MLE, but less efficient: use the distribution if known.

Quasi-deviance: saturated $Q = 0$, so scaled $D_Q/\phi = -2\sum Q_i$, $D_Q = -2\sum_i\int_{y_i}^{\mu_i}\frac{y_i-t}{v(t)}dt$.

Quasi-Poisson: $v = \mu$, $Q_i = \frac{1}{\phi}[y_i\log\mu_i - \mu_i]$ + const. **Quasi-binomial:** $v = \mu(m-\mu)/m$, $Q_i = \frac{1}{\phi}[y_i\log\frac{\mu_i}{m_i-\mu_i} + m_i\log(m_i-\mu_i)]$ + const. ϕ cancels in IWLS, so **same** $\hat{\beta}$, $\hat{\mu}$, D as Poisson/binomial.

Numeric Q_i (definite integral, ≤ 0): $v = \mu$: $\phi Q_i = y\log\frac{\mu}{y} - (\mu - y)$. $v = \mu(m-\mu)/m$: $y\log\frac{\mu}{y} + (m-y)\log\frac{m-\mu}{m-y}$. $v = \mu^2$: $1 - \frac{y}{\mu} + \log\frac{y}{\mu}$. $v = \mu^3$: $-\frac{(y-\mu)^2}{2y\mu^2}$.

SE: $\mathrm{se}_Q = \sqrt{\hat{\phi}}\,\mathrm{se}_P$, $\hat{\phi}$ of that same model; reverse: $\mathrm{se}_P = \mathrm{se}_Q/\sqrt{\hat{\phi}}$. Estimate never changes.

Coefficient: $t^* = \hat{\beta}/\mathrm{se}_Q \approx t_{n-p}$; CI $\hat{\beta} \pm t_{n-p}(.975)\mathrm{se}_Q$ (1.96 if no t given), exp for ratio.

Nested F: $F = \frac{(D_A - D_B)/s}{\hat{\phi}} \approx F_{s,n-p_B}$, $\hat{\phi}$ from the BIG model B. No adequacy test, no χ^2, AIC is NA (no likelihood), `step`/`drop1` do not work. Exact for Normal (LM F test), approximate otherwise.

New data type: choose link and v from mean-variance (e.g. proportions: logit, $\mu(1-\mu)$), fit IWLS, $\hat{\phi}$ from Pearson. $\hat{\phi} < 1$ (underdispersion) is allowed. iid mean λ, var $\phi\lambda$: $\hat{\lambda} = \bar{y}$, $\hat{\phi} = \frac{\sum(y_i-\bar{y})^2}{\bar{y}(n-1)}$.

Degrees of freedom: null $n-1$; residual, GOF χ^2, Wald t: $n-p$; LRT χ^2: $s =$ difference in residual df; quasi F: $(s, n-p_B)$; factor: $L-1$; interaction of two factors: $(L_1-1)(L_2-1)$.

Maid Mast Codes PT1: GLM

`glm(cbind(y,m-y)~x, family=binomial, d)`: grouped binomial. `family=poisson, quasipoisson, quasibinomial, Gamma(link="log")`. `subset=-25` drops row 25.

- **Estimate** $\hat{\beta}_j$; **Std. Error** $\sqrt{[I(\hat{\beta})^{-1}]_{jj}}$; **z value** = Est/SE; **Pr(>|z|)** two-sided. t if ϕ estimated.
- **Dispersion:** 1 or $X^2/(n-p)$. **Null deviance** on $n-1$; **Residual deviance** on $n-p$. **AIC**. **Fisher Scoring iterations:** IWLS steps.
- `anova(A,B,test="Chisq")`: Resid.Df, Resid.Dev, Df = s, Deviance = ΔD. `test="F"` on quasi or Gamma fits adds F (plain Poisson/binomial: $\Delta D/s$, ignores overdispersion). `anova(m,test="Chi")`: terms added in order.
- `anova(A,B,dispersion=phi,test="F")` on binomial fits: same F as quasi, but p uses $F_{s,\infty}$.
- `step(m)`: each row is the model after that change; drop the term with lowest AIC if below `<none>`; stops when `<none>` lowest.
- `sum(residuals(m,type="pearson")^2)/m$df.residual` = $\hat{\phi}$.
- `pchisq(D,df,lower.tail=FALSE)` = $P(\chi^2_{df} > D)$.
- `predict(m,newdata=list(x=1),type="response")`: $\hat{\mu}$ (`cbind` binomial: $\hat{p}$, multiply by m; same for `fitted`; offset: newdata needs t). `type="link"`, `se.fit=TRUE`: $\hat{\eta}$, se. `rstudent` jack-knife (exact if $\phi = 1$); `influence(m)$hat`; `cooks.distance`; `halfnorm`.
- `optim(start,logL,control=list(fnscale=-1))` maximises l; start values matter.
- `summary(m,dispersion=phi)` rescales SEs. Old papers: `drop1(m,scale=phi,test="F")`, its F uses $D/(n-p)$, not Pearson.
- `vcov(m)` = V (quasi includes $\hat{\phi}$); `rstandard(m)` = r_{SD}; `hatvalues(m)` = H_{ii}.
- `z<-predict(m)+(y-mu)/mu`: linearised response, log link.
- `dnorm(x,mean,sd)`, `rnorm(n,mean,sd)` take sd. `rgamma(n,shape,rate)`.

Interpretation lines: **Log link:** "Holding other predictors fixed, a 1-unit rise in x multiplies the expected count by $e^{\hat{\beta}}$, a $100(e^{\hat{\beta}}-1)$% change." **Logit:** same for the odds, not the probability. **Identity:** adds $\hat{\beta}$. **Inverse:** $1/\mu$ rises by $\hat{\beta}$, so $\hat{\beta} > 0$ lowers μ. **Factor:** "relative to the reference level". $\log x$ **with log link:** doubling x multiplies μ by $2^{\hat{\beta}}$. **Interaction:** the effect of one predictor depends on the other; a cell vs baseline is exp(its main effects + interaction); add the intercept for the cell mean.

Hard subquestions (R output):

- **Which model, 2 quantitative reasons:** smaller AIC, and $\Delta D > \chi^2_s(.95)$ rejects the smaller one.
- **2 reasons overdispersed:** $D \gg df$ and $\hat{\phi} \gg 1$.
- **Why F, t not χ^2, z:** ϕ unknown, estimated by $\hat{\phi}$.
- **Poisson vs quasi-Poisson:** Poisson fixes $\mathrm{Var}Y = \mu$. Quasi keeps the same mean model, sets $\mathrm{Var}Y = \phi\mu$ with ϕ estimated from Pearson and assumes no distribution. Same $\hat{\beta}$; SEs $\times\sqrt{\hat{\phi}}$; tests become t, F; no AIC.
- **Interaction tested twice:** z from Fisher information $\sim \mathrm{N}(0,1)$, and $\Delta D \sim \chi^2_1$. Prefer ΔD.
- **Linear assumption evidence:** $\log(y/t)$ vs x straight; interaction plot parallel lines = no interaction.
- **Outlier:** clear point in half-normal Cook/jack-knife plot. Refit, re-check $\hat{\phi}$: overdispersion typically remains.

Mixture model: (1) $Z_i \sim \text{Categorical}(\pi_1,\dots,\pi_K)$, Z_i latent label, π_k mixture proportions. (2) $X_i|Z_i = k \sim P(X_i|\Theta_k)$ component. $P(x_i) = \sum_k \pi_k f(x_i;\Theta_k)$, $L(\theta) = \prod_i \sum_k \pi_k f(x_i;\Theta_k)$. Free parameters: $K-1$ proportions + component parameters. A mixture is not a linear combination: $\pi_a X_a + \pi_b X_b$ of Normals is Normal; a mixture may be bimodal, or unimodal and skewed.
Mean: $\bar\mu = \sum \pi_k \mu_k$. **Var:** $\sum \pi_k \sigma_k^2 + \sum \pi_k (\mu_k - \bar\mu)^2$. Poisson mixtures are overdispersed.
Clustering: $P(Z_i = k|X_i) = \frac{\pi_k f(x_i;\Theta_k)}{\sum_j \pi_j f(x_i;\Theta_j)}$; assign to the largest. Common factors cancel ($x!$, $\sqrt{2\pi}$) but not π_k or unequal $1/\sigma_k$. Poisson, $K=2$: $P_{i1} = 1/[1 + \frac{\pi_2}{\pi_1}(\frac{\lambda_2}{\lambda_1})^{x_i} e^{-(\lambda_2-\lambda_1)}]$. Normal, equal σ^2, $\mu_1 < \mu_2$: cluster 1 if $x < \frac{\mu_1+\mu_2}{2} + \frac{\sigma^2 \log(\pi_1/\pi_2)}{\mu_2-\mu_1}$.

EM algorithm: $l(\theta) = \sum_i \log \sum_k \pi_k f$: the sum inside the log blocks a closed form. If Z were known it would be easy.
Incomplete log-lik: $\log P(X|\theta) = \log \sum_Z P(X,Z|\theta)$. **Complete:** $\log P(X,Z|\theta)$. **Q:** $Q(\theta,\theta^0) = \mathbb{E}_{Z|X,\theta^0}[\log P(X,Z|\theta)]$.
Steps: (1) Set θ^0; evaluate incomplete log-lik. (2) **E:** compute $P(Z|X,\theta^0)$. (3) **M:** $\theta = \arg\max_\theta Q(\theta,\theta^0)$. (4) Evaluate incomplete log-lik; stop if change $< \epsilon$, else $\theta^0 \leftarrow \theta$, go to 2.
Deriving Q: write $P_{ik} = P(Z_i = k|X_i,\theta^0)$.
$P(X,Z|\theta) = \prod_i \prod_k [\pi_k f(x_i;\Theta_k)]^{I(Z_i=k)}$
$\log P(X,Z|\theta) = \sum_i \sum_k I(Z_i=k)[\log \pi_k + \log f(x_i;\Theta_k)]$
$\mathbb{E}[I(Z_i=k)] = P_{ik}$ and linear in I so random, so
$Q = \sum_i \sum_k P_{ik}[\log \pi_k + \log f(x_i;\Theta_k)]$
$K=2$: $\pi_1 = \pi$, $\pi_2 = 1-\pi$. Normal: $\log f = -\frac{1}{2}\log(2\pi\sigma_k^2) - \frac{(x_i-\mu_k)^2}{2\sigma_k^2}$ (that π is 3.14). Poisson: $x_i \log\lambda_k - \lambda_k - \log x_i!$. Normal answer in full:
$Q = \sum_i \sum_{k=1}^2 P_{ik}[\log\pi_k - \frac{1}{2}\log 2\pi\sigma_k^2 - \frac{(x_i-\mu_k)^2}{2\sigma_k^2}]$
M-step, $N_k = \sum_i P_{ik}$ (effective count):
- π: $\frac{\partial Q}{\partial \pi} = \frac{N_1}{\pi} - \frac{N_2}{1-\pi} = 0$, so $\hat\pi_k = N_k/n$. $K \geq 3$: Lagrange $\sum \pi_k \log \pi_k + \nu(1 - \sum \pi_k)$ gives $\pi_k = N_k/\nu$; sum over k: $\nu = n$.
- **Normal:** $\hat\mu_k = \sum_i P_{ik} x_i / N_k$, $\hat\sigma_k^2 = \sum_i P_{ik}(x_i - \hat\mu_k)^2 / N_k$ (from $\frac{\partial Q}{\partial \sigma^2} = \sum P_{ik}[-\frac{1}{2\sigma^2} + \frac{(x_i-\mu)^2}{2\sigma^4}] = 0$). Mean 0: $\sum P_{ik} x_i^2 / N_k$. Shared σ^2: $\sum_i \sum_k P_{ik}(x_i - \hat\mu_k)^2 / n$.
- **Poisson:** $\hat\lambda_k = \sum P_{ik} x_i / N_k$; $Q'' = -\sum P_{ik} x_i/\lambda_k^2 < 0$. **Bin**$(m, p_k)$: $\hat p_k = \sum P_{ik} x_i / (m N_k)$. **Exp(rate λk):** $\hat\lambda_k = N_k / \sum P_{ik} x_i$.

Twists rule: collect every term with the parameter across all components and blocks, differentiate once. Exp(λ), Exp(2λ): $\hat\lambda = n/\sum(P_{i1} + 2P_{i2})x_i$. Exposure t_i: $\hat\lambda_k = \sum P_{ik} x_i / \sum P_{ik} t_i$. N($\mu$,$\sigma^2$), N($\mu$,9$\sigma^2$): $w_i = P_{i1} + P_{i2}/9$, $\hat\mu = \sum w_i x_i / \sum w_i$, $\hat\sigma^2 = \sum w_i (x_i - \hat\mu)^2 / n$.
n' points known from component j (not drawn from the mixture): no Z, so they add $\sum \log f_j(x_i)$ to Q with weight 1 and no π. $\hat\pi_k = N_k/n$, unlabelled n only. $\hat\mu_j = (\sum P_{ik} x_i + \sum_{lab} x_i)/(N_j + n')$; same for σ_j^2 with $(x_i - \hat\mu_j)^2$. Incomplete log-lik: $\sum_{unlab} \log \sum_k \pi_k f_k + \sum_{lab} \log f_j$.
Observed labels from the mixture: count in both: $\hat\pi_k = (N_k + m_k)/(n+m)$. **Known parameters:** do not differentiate. **Tied:** N(μ,1),N($-\mu$,1): differentiate once, $\hat\mu = \sum(P_{i1} - P_{i2})x_i/n$. **Zero-inflated:** $P_{i1} = 0$ if $x_i > 0$; $\hat\lambda = \sum x_i/(n - N_1)$. $K=3$: $\pi_3 = 1 - \pi_1 - \pi_2$, three terms in the denominator.
Behaviour: incomplete log-lik **never decreases** (a drop means a bug). Local maxima: sensitive to initial values, so run many starts, keep the **highest incomplete log-lik**. **Label switching:** same log-lik, permuted labels, same fit. Equal starts ($\lambda_1^0 = \lambda_2^0$) get stuck at $\hat\lambda_k = \bar x$, π unchanged. Overlap: P_{ik} less decisive, EM slower and more start-sensitive. Do not compare log-lik across different data.
By hand: table of x_i, P_{i1}, P_{i2}; check $N_1 + N_2 = n$. M-step is the complete-data MLE with hard counts replaced by weights P_{ik}. To choose between runs, plug each $\hat\theta$ into $\sum_i \log(\pi f_1(x_i) + (1-\pi) f_2(x_i))$. Keep $\log x_i!$ when reporting Q or log-lik values.
Proof (Jensen): $q(Z) = P(Z|X,\theta^0)$:
$l(\theta) = \log \sum_Z q \frac{P(X,Z|\theta)}{q} \geq \sum_Z q \log \frac{P(X,Z|\theta)}{q}$
$= Q(\theta,\theta^0) + C$, $C = -\sum q \log q$. Equality at θ^0, as $P(X,Z|\theta^0)/q = P(X|\theta^0)$ is constant. M-step: $Q(\hat\theta,\theta^0) \geq Q(\theta^0,\theta^0)$. So $l(\hat\theta) \geq Q(\hat\theta,\theta^0) + C \geq Q(\theta^0,\theta^0) + C = l(\theta^0)$.

Maid Mast Codes PT2: EM
- `Z=sample(1:K,n,replace=TRUE,prob=pi); X=rnorm(n,mu[Z],sd[Z])`: simulate.
- `L[,k]=dnorm(X,mu[k],s)*w[k]`: $n \times K$ matrix $P(X_i, Z_i{=}k)$.
- `P_ik=prob.x.z/rowSums(prob.x.z)`: E-step.
- `w.new=colSums(P_ik)/sum(P_ik)`: $\hat\pi_k$; `sum(P_ik)` $= n$.
- `mu.new=colSums(P_ik*X)/colSums(P_ik)`: $\hat\mu_k$.
- `ill=sum(log(rowSums(prob.x.z)))`: incomplete log-lik.
- `delta.ll`, `epsilon`, `max.iter`: hitting `max.iter` means not yet converged; `length(log_liks)` = iterations +1; negative `delta.ll` also stops the loop, so a bug can look converged. Bugs: `/colSums` in E-step; `sum(log(prob.x.z))`.

Bayes: Frequentist: θ fixed, probability = long-run frequency. Bayesian: θ random, probability = degree of belief.
$p(\theta|x) = \frac{f(x|\theta)p(\theta)}{f(x)} \propto f(x|\theta)\,p(\theta)$
Likelihood × prior. Marginal $f(x) = \int f(x|\theta)p(\theta)d\theta$. **Point estimate:** posterior mean (minimises squared error loss). **Credible interval:** central 95% of the posterior, a direct probability statement about θ. Normal posterior: mean ± 1.96 sd; samples: `quantile(th,c(.025,.975))`.

Kernels: **Gamma**(a,b): $\tau^{a-1}e^{-b\tau}$.
InvGamma(a,b): $x^{-a-1}e^{-b/x}$. **Normal:** $e^{-\frac{A}{2}\mu^2 + B\mu}$ is N(B/A, $1/A$). **Beta**(a,b): $\theta^{a-1}(1-\theta)^{b-1}$.
ln λ: Poisson likelihood is $\lambda^{\sum y_i} e^{-n\lambda}$, Exp is $\lambda^n e^{-\lambda\sum y_i}$: Gamma kernels. Exp(rate b) = Gamma$(1,b)$.
Recipe: joint → keep factors with the parameter → add exponents → name the kernel. Summaries: $\sum(x_i - c)^2 = \sum(x_i - \bar x)^2 + n(\bar x - c)^2$. $\tau^{n/2}$ is constant for μ but not for τ: keep it. Each prior appears once.

Likelihood (n)	Prior	Posterior
Bin(m,θ)	Beta	Beta$(a+\sum y_i,\ b+nm-\sum y_i)$
Geom ($x \geq 1$)	Beta	Beta$(a+n,\ b+\sum y_i - n)$
Poisson(λ)	Gamma	Gamma$(a+\sum y_i,\ b+n)$
Exp(rate λ)	Gamma	Gamma$(a+n,\ b+\sum y_i)$
N$(\mu_0, 1/\tau)$	Gamma	Gamma$(a+\frac{n}{2},\ b+\frac{1}{2}S)$
N(μ_0,σ^2)	InvG	InvGamma$(a+\frac{n}{2},\ b+\frac{1}{2}S)$

$S = \sum(x_i - \mu_0)^2$. Poisson$(t_i\lambda)$: rate $b + \sum t_i$.
Normal mean, σ^2 known: prior N$(\mu, \sigma^2/n_0)$ (n_0 = implicit sample size):
$\theta|x \sim \text{N}\left(\frac{n_0\mu + n\bar x}{n_0+n},\ \frac{\sigma^2}{n_0+n}\right)$
Prior N(m, s^2): precisions add: variance $(\frac{1}{s^2} + \frac{n}{\sigma^2})^{-1}$, mean = variance $\times(\frac{m}{s^2} + \frac{n\bar x}{\sigma^2})$. $\bar X \sim$ N$(m, \frac{\sigma^2}{n} + s^2)$ marginally.
Posterior mean is a precision-weighted average of prior mean and MLE. Variance is below both prior and MLE variance. $n \to \infty$: $\to$ N$(\bar x, \sigma^2/n)$, prior forgotten. $n_0 \to 0$: flat prior.
Posterior predictive: $p(\tilde x|x) = \int p(\tilde x|\theta)p(\theta|x)d\theta$. Normal: N(post. mean, post. var + σ^2). Simulate: $\theta^{(l)} \sim p(\theta|x)$, then $\tilde x^{(l)} \sim p(\tilde x|\theta^{(l)})$; average. Known mean μ_0, $\tau|x \sim$ Gamma(a^*, b^*): 3-parameter t with $\nu = 2a^*$, location μ_0, scale^2 $s^2 = b^*/a^*$, pdf $\propto (1 + \frac{(x-\mu_0)^2}{\nu s^2})^{-(\nu+1)/2}$.
Pseudo-counts: Beta(a,b) acts like $a+b$ prior trials; Beta(1,1) = U(0,1). Poisson-Gamma mean: $\frac{a+\sum y_i}{b+n} = w\frac{a}{b} + (1-w)\bar y$, $w = \frac{b}{b+n}$.
Marginals: Beta-binomial $P(X{=}x) = \binom{m}{x} B(a+x, b+m-x)/B(a,b)$; Poisson-Gamma gives NegBin.
Priors: vague N$(0, 10^4)$, Gamma(.001,.001); **informative**: concentrated. **Improper:** does not integrate to 1, $p(\mu) \propto 1$, $p(\tau) \propto 1/\tau$ (Gibbs: $a = b = 0$); fine if the posterior is proper ($p(\lambda) \propto 1/\lambda$, Poisson: Gamma$(\sum y_i, n)$, needs $\sum y_i > 0$). $p(\mu, \sigma^2) \propto 1/\sigma^2$ goes straight into π in MH.
Spike prior: γ discrete: $p(\gamma|x) \propto p(x|\gamma)p(\gamma)$, with $p(x|\gamma) = \int p(x|\mu)p(\mu|\gamma)d\mu$; sample γ, then $\mu|\gamma, x$.
Conjugate exp family: $f = \alpha(x)e^{x\theta + \beta(\theta)}$, prior $\propto e^{\gamma\theta + \delta\beta(\theta)}$: posterior $(\gamma + x, \delta + 1)$.

Simulation: **Monte Carlo:** $\frac{1}{N}\sum g(X_j) \approx \mathbb{E}g(X)$; proportions estimate probabilities. `set.seed` makes it repeatable.
Discrete: $U \sim$ U(0,1); `x<-0; while(F(x)<U) x<-x+1`. Ends with $F(x-1) < U \leq F(x)$, so $P(X = x) = F(x) - F(x-1) = p(x)$. **Read a pmf from a loop:** `while(U>G(Y)) Y<-Y+1` means $F = G$, then difference it. `sample(x,n,replace=TRUE,prob=p)`. Expected comparisons: $\sum$ (tests needed) × prob, so test likely values first.
Inversion: $Y = F^{-1}(U)$: $P(Y \leq y) = P(U \leq F(y)) = F(y)$. Needs F^{-1}. Exp(λ): $-\log(1-U)/\lambda$; U(a,b): $a + (b-a)U$; logistic: $\log \frac{U}{1-U}$; Weibull: $(-\log U)^{1/k}/\lambda$.
Rejection, uniform: $f \leq k$ on $[a,b]$. (1) $X \sim$ U(a,b), $Y \sim$ U$(0,k)$. (2) If $Y < f(X)$ return X, else go to 1.
General: envelope density h, $k \geq \sup_x f(x)/h(x)$. (1) $X \sim h$. (2) $Y \sim$ U$(0, kh(X))$. (3) If $Y < f(X)$ return X, else go to 1. Works with unnormalised f^* (posterior): $k \geq \sup f^*/h$. Prior as h: $k = \max_\theta p(x|\theta)$, keep if $U \leq p(x|\theta)/k$.
Why: (X,Y) is uniform under kh (density $1/k$); points kept are uniform under f, so $X \sim f$.
Derive output pdf: $P(X \leq x|\text{accept}) = P(X \leq x, \text{accept})/P(\text{accept}) = \int_{-\infty}^x h(t)\frac{f(t)}{kh(t)}dt / \frac{1}{k} = F(x)$.
Efficiency: expected candidates per sample. $P(\text{accept}) = 1/k$ (with f^*: $\int f^*/k$). $N \sim$ Geom$(1/k)$, $\mathbb{E}N = k$ (f^*: $k/\int f^*$). Least rejection: smallest valid k, h close to f. Uniform envelope: rectangle height $\sup f$ on $[a,b]$, so $k = (b-a)\sup f$.
Gamma(shape m, rate λ), $m > 1$, $h = \mu e^{-\mu x}$: f/h peaks at $x = \frac{m-1}{\lambda - \mu}$. Best $\mu = \lambda/m$, $k^* = m^m e^{-(m-1)}/\Gamma(m)$. $k^* = \infty$ if $m < 1$ or $\lambda \leq \mu$.

Maid Mast Codes PT3: Simulation
- `x<-runif(1,a,b); y<-runif(1,0,K); if(y<fx(x)) return(x)`: rejection.
- `X<- -log(runif(1))*m/lambda`: Exp(rate λ/m) by inversion.
- `tha<-rbeta(N,2,3); X<-rbinom(N,10,tha); table(X)/N`: marginal pmf by simulating parameter then data.
- `while(runif(1)>p) Y<-Y+1` from $Y = 0$: failures before first success, $P(Y = y) = p(1-p)^y$.

MCMC: build a Markov chain whose stationary distribution is the posterior. Long-run draws are dependent but approximately from the posterior.
Metropolis-Hastings: target $\pi \propto p(\theta)p(x|\theta)$ (constant not needed). **Proposal** $q(\theta,\theta')$: density of new θ' given current θ; must be simulable.
(1) Start $\theta^{(0)}$, $t = 0$. (2) Propose $\theta' \sim q(\theta^{(t)}, \cdot)$. (3)
$A = \min\left(1,\ \frac{\pi(\theta')\,q(\theta',\theta)}{\pi(\theta)\,q(\theta,\theta')}\right)$
(4) With probability A set $\theta^{(t+1)} = \theta'$, else $\theta^{(t+1)} = \theta$ ($U \leq A$). (5) $t \leftarrow t+1$, go to 2.
Random walk: $\theta' = \theta +$ symmetric noise (U$[-\epsilon,\epsilon]$, N$(0,\epsilon^2)$): q cancels, $A = \min(1, \pi(\theta')/\pi(\theta))$.
Uphill always accepted, downhill sometimes, so time spent $\propto \pi$. **Independence:** $q(\theta,\theta') = q(\theta')$, so $A = \min\{1, w(\theta')/w(\theta)\}$, $w = \pi/q$. **Walk on** $\log\tau$: extra factor τ'/τ.
Numeric A recipe: (i) π = likelihood × prior at new and current. (ii) q ratio: top is density of proposing current from new, bottom new from current. Multi-part proposals multiply: $\tau_n \sim g(\cdot|\tau_c)$, $\mu_n \sim h(\cdot|\mu_c,\tau_n)$ gives $q(c,n) = g(\tau_n|\tau_c)h(\mu_n|\mu_c,\tau_n)$, $q(n,c) = g(\tau_c|\tau_n)h(\mu_c|\mu_n,\tau_c)$. (iii) Cancel common constants only. (iv) min(1,·). Out of support: $\pi = 0$, $A = 0$. Log form (symmetric q): `A<-min(1,exp(lp(new)-lp(cur)))`; else add log q ratio.
Tuning: want large moves and high acceptance; they compete, so aim for intermediate A. Tiny steps: high acceptance, slow crawl. Huge steps: low acceptance, sticky. Both: high autocorrelation, low ESS.
Read π off A: symmetric q and $A = \min(1, r(\theta_n)/r(\theta_o))$ give $\pi \propto r$.
Why it works (not examinable): detailed balance $\pi(\theta)p(\theta \to \theta') = \pi(\theta')p(\theta' \to \theta)$ makes π stationary; aperiodic, irreducible: limiting; ergodic: averages converge.

Gibbs sampler: needs only full conditionals. $\theta = (\theta_1,\dots,\theta_k)$. Start $\theta^{(0)}$. Each sweep: $\theta_j^{(t+1)} \sim p(\theta_j|x, \text{latest values of others})$, $j = 1,\dots,k$. Discard burn-in m; keep the rest as a dependent sample; marginals: keep that coordinate.
Use output: $\mathbb{E}[g(\theta)|x] \approx$ average of $g(\theta^{(t)})$ after burn-in; $P(\theta > c|x) \approx$ proportion above c; credible interval from sample quantiles.
Special case of MH: proposal = full conditional, so $A = 1$. **Limit:** strong correlation between coordinates gives tiny moves and slow exploration.
Not a named distribution? Finite support: sample with weights (R normalises `prob`). Otherwise an MH step for that coordinate.
Standard conditionals ($x_i \sim$ N$(\mu, 1/\tau)$, $S = \sum(x_i - \mu)^2$):
- $\mu \sim$ N$(m, \frac{1}{\kappa})$: $\mu|\tau, x \sim$ N$(\frac{\kappa m + n\tau\bar x}{\kappa + n\tau}, \frac{1}{\kappa + n\tau})$.
- $\mu|\tau \sim$ N$(m, \frac{1}{\kappa\tau})$: $\mu|\tau, x \sim$ N$(\frac{\kappa m + n\bar x}{\kappa + n}, \frac{1}{(\kappa+n)\tau})$.
- $\tau \sim$ Gamma(a,b): $\tau|\mu, x \sim$ Gamma$(a + \frac{n}{2}, b + \frac{S}{2})$; with the $\mu|\tau$ prior: Gamma$(a + \frac{n+1}{2}, b + \frac{S}{2} + \frac{\kappa}{2}(\mu - m)^2)$.
- $\sigma^2 \sim$ InvGamma(a,b): InvGamma$(a + \frac{n}{2}, b + \frac{S}{2})$; prior $\mu|\sigma^2 \sim$ N$(m, \sigma^2/\kappa)$: InvGamma$(a + \frac{n+1}{2}, b + \frac{S}{2} + \frac{\kappa}{2}(\mu - m)^2)$.
- **Two samples** $\bar x$ (n_1), $\bar y$ (n_2), common τ: Gamma$(a + \frac{n_1+n_2}{2}, b + \frac{S_x+S_y}{2})$; μ_1 uses x only.
- **One μ, two sources:** $x_i \sim$ N$(\mu, \frac{1}{\tau})$ (n), $y_j \sim$ N$(c\mu, \frac{1}{d\tau})$ (m), $\mu \sim$ N$(\mu_0, \frac{1}{\kappa})$: $A = \kappa + (n + c^2 dm)\tau$, $B = \kappa\mu_0 + \tau(\sum x_i + cd\sum y_j)$; $\tau|\cdot \sim$ Gamma$(a + \frac{n+m}{2}, b + \frac{1}{2}[\sum(x_i - \mu)^2 + d\sum(y_j - c\mu)^2])$.
- **Hierarchical:** $y_{ij} \sim$ N$(\theta_j, \frac{1}{\tau})$, $\theta_j \sim$ N$(\mu, \frac{1}{\tau_\theta})$, $\mu \sim$ N$(m, \frac{1}{\kappa})$: $\theta_j|\cdot$ precision $n_j\tau + \tau_\theta$, mean $(n_j\tau\bar y_j + \tau_\theta\mu)$/prec; $\mu|\cdot$ precision $\kappa + J\tau_\theta$, mean $(\kappa m + \tau_\theta\sum\theta_j)$/prec (no y); $\tau_\theta|\cdot \sim$ Gamma$(a + \frac{J}{2}, b + \frac{1}{2}\sum(\theta_j - \mu)^2)$.
- **Bivariate prior,** $\Lambda = \Sigma^{-1}$, zero mean, $x_i \sim$ N(μ_1, σ^2): $-\frac{1}{2}(\Lambda_{11}\mu_1^2 + 2\Lambda_{12}\mu_1\mu_2 + \Lambda_{22}\mu_2^2)$ in the exponent gives $A = \Lambda_{11} + n/\sigma^2$, $B = n\bar x/\sigma^2 - \Lambda_{12}\mu_2$: $\mu_1|\mu_2, x \sim$ N$(B/A, 1/A)$.
- **Change point:** $Y_i \sim$ Pois(λ_1), $i \leq m$; Pois(λ_2), $i > m$; Gamma priors; m uniform on $1..n$: $\lambda_1|\cdot \sim$ Gamma$(\alpha_1 + \sum_1^m y_i, \beta_1 + m)$; $\lambda_2|\cdot \sim$ Gamma$(\alpha_2 + \sum_{m+1}^n y_i, \beta_2 + n - m)$; $p(m|\cdot) \propto (\lambda_1/\lambda_2)^{\sum_1^m y_i} e^{(\lambda_2 - \lambda_1)m}$.
- **Hidden count:** $X|p, N \sim$ bin(N,p), $p \sim$ Beta(a,b), $N \sim$ Pois(λ): $p|\cdot \sim$ Beta$(x + a, N - x + b)$; $N - x|\cdot \sim$ Pois$(\lambda(1-p))$.
- **Discrete** $\mu \in \{0,1\}$: $P(\mu = 1|\theta) = f(\theta,1)/[f(\theta,0) + f(\theta,1)]$.

MCMC diagnostics: (1) Converged? (2) How correlated?
Trace plots: run several chains from **overdispersed** starts (more variable than the target). Good: chains overlap, stable, "hairy caterpillar". Separate or drifting: not converged. Slow wander, "woolly snake": high autocorrelation. **Burn-in:** discard early iterations affected by the start; read length off the trace.
BGR: M chains, each length $L + T$; keep last T. **B** = sd of the pooled MT draws (between). **W** = average of the M within-chain sds (within). $\hat R = B/W < 1.05$: accept last T as stationary and pool. $\hat R \gg 1$: not converged. $T \to \infty$: $\hat R \to 1$.
Trace vs step size: tiny steps: smooth wander; huge steps: flat runs then jumps; good: fuzzy band.
Not converged: run longer; better proposal. Passing is no proof if important regions were missed by the starts.
Autocorrelation: lag k: $\rho_k = \text{cor}(\theta^{(t)}, \theta^{(t+k)})$; `acf(chain)`; `cor(chain[-n],chain[-1])` is $\hat\rho_1$. **Thinning:** keep every kth; simple, saves memory, throws away information.
ESS: independent sample size with the same information: $n/(1 + 2\sum_i \rho_i) \approx n\frac{1-\rho_1}{1+\rho_1}$. Aim ESS > 4000. E.g. $n = 5000$, $\rho_1 = .9$: ESS ≈ 263.
Accuracy: $\text{Var}(\bar h) \approx \frac{\sigma^2}{n}(1 + 2\sum\rho_i)$: iid SE is too small. MC s.e.: iid $s/\sqrt{N}$; proportion $\sqrt{\hat p(1-\hat p)/N}$; chain $s/\sqrt{\text{ESS}}$. Posterior sd does not shrink with N.
Blocking: Q blocks, MC error = sd(block means)$/\sqrt{Q}$; rule: $\leq 5\%$ of posterior sd.

Variational inference: turns inference into optimisation: find $q^* \in D$ nearest the posterior. Faster for large data and complex models, at a cost in accuracy.
KL: $KL(q\|p(\theta|y)) = \int q(\theta)[\log q(\theta) - \log p(\theta|y)]d\theta \geq 0$. Not symmetric. Penalises q for mass where p is tiny, not for missing mass, so q sits inside p.
ELBO: $p(\theta|y)$ unknown, but $KL = -ELBO + \log p(y)$ with
$ELBO(q) = -\int q[\log q - \log p(\theta)p(y|\theta)]d\theta$
$= \mathbb{E}_q[\log p(\theta,y)] - \mathbb{E}_q[\log q]$. $\log p(y)$ is free of q, so **min KL = max ELBO**; $KL \geq 0$ gives $\log p(y) \geq ELBO$.
Mean field: $q(\theta) = \prod_j q_j(\theta_j)$: parameters independent under q. An assumption on the approximation, not on the model.
CAVI: update one factor at a time, others fixed: $q_j^*(\theta_j) \propto \exp\{\mathbb{E}_{-j}[\log p(\theta,y)]\}$. ELBO rises to a local optimum (a drop means a bug). Stop when ELBO change is small. Sensitive to initial values: run several, keep highest ELBO. It updates distributions; it draws no samples.
Recipe: (1) $\log p(\theta,y)$ = log likelihoods + log priors. (2) Keep terms with θ_j. (3) Take $\mathbb{E}$ over the others (linear). (4) Collect powers, name the kernel. Same kernel as Gibbs, with terms in other parameters replaced by their expectations.
Use: $\mathbb{E}(x-\mu)^2 = (x - \mathbb{E}\mu)^2 + V\mu = \sigma^{2*} + \mu^{*2} - 2x\mu^* + x^2$; $\mathbb{E}\mu^2 = (\mathbb{E}\mu)^2 + V\mu$; $\mathbb{E}\tau = a^*/b^*$; $\mathbb{E}\log\tau = \psi(a^*) - \log b^*$; $\mathbb{E}[1/\sigma^2] = a^*/b^*$.
Normal-Gamma: $x_i \sim$ N$(\mu, \frac{1}{\tau})$, $\mu|\tau \sim$ N$(\mu_0, \frac{1}{\lambda_0\tau})$, $\tau \sim$ Gamma(a_0, b_0):
$q_\mu^* =$ N(μ^*, σ^{2*}): $\mu^* = \frac{\lambda_0\mu_0 + n\bar x}{\lambda_0 + n}$, $\sigma^{2*} = \frac{1}{(\lambda_0+n)\mathbb{E}\tau}$.
$q_\tau^* =$ Gamma(a^*, b^*): $a^* = a_0 + \frac{n+1}{2}$, $b^* = b_0 + \frac{1}{2}\sum_i \mathbb{E}(x_i - \mu)^2 + \frac{\lambda_0}{2}\mathbb{E}(\mu - \mu_0)^2$.
Only σ^{2*} and b^* change between iterations.
Variance form $\sigma^2 \sim$ InvGamma(a_0, b_0), $\mu|\sigma^2 \sim$ N$(\mu_0, \sigma^2/\lambda_0)$: $q_{\sigma^2}^*$ is InvGamma with the same two parameters.
Independent priors $\mu \sim$ N$(m, \frac{1}{\kappa})$, $\tau \sim$ Gamma(a,b): $q_\mu^* =$ N$(\frac{\kappa m + n\bar x\mathbb{E}\tau}{\kappa + n\mathbb{E}\tau}, \frac{1}{\kappa + n\mathbb{E}\tau})$; $q_\tau^* =$ Gamma$(a + \frac{n}{2},\ b + \frac{1}{2}[\sum(x_i - \mathbb{E}\mu)^2 + nV\mu])$. Shape $a + \frac{n}{2}$, not $\frac{n+1}{2}$; here both factors move each round.
Exact: $\mu|\tau, x \sim$ N$(\frac{\lambda_0\mu_0 + n\bar x}{\lambda_0 + n}, \frac{1}{(\lambda_0+n)\tau})$, $\tau|x \sim$ Gamma$(a_0 + \frac{n}{2}, b_0 + \frac{1}{2}\sum(x_i - \bar x)^2 + \frac{n\lambda_0(\bar x - \mu_0)^2}{2(\lambda_0+n)})$.
ELBO here: $\mathbb{E}[\log p(x,\mu,\tau)] - \mathbb{E}\log q_\mu - \mathbb{E}\log q_\tau$, with $\mathbb{E}\log q_\mu = -\frac{1}{2}\log\sigma^{2*} + c$, $\mathbb{E}\log q_\tau = \log b^* - \log\Gamma(a^*) + (a^* - 1)\psi(a^*) - a^*$.
Bivariate Normal target: $\Lambda = \Sigma^{-1}$: $q_1^* =$ N$(\mu_1 - \frac{\Lambda_{12}}{\Lambda_{11}}(\mathbb{E}x_2 - \mu_2), \frac{1}{\Lambda_{11}})$. $1/\Lambda_{11} = \sigma_1^2 - \sigma_{12}^2/\sigma_2^2 < \sigma_1^2$: the conditional variance. Means right, but VI underestimates variance and ignores correlation; worse as correlation rises. ELBO up to a constant, $q_i =$ N(m_i, s_i^2): $-\frac{1}{2}[\Lambda_{11}(s_1^2 + (m_1 - \mu_1)^2) + \Lambda_{22}(s_2^2 + (m_2 - \mu_2)^2) + 2\Lambda_{12}(m_1 - \mu_1)(m_2 - \mu_2)] + \frac{1}{2}\log s_1^2 + \frac{1}{2}\log s_2^2$.
Bivariate prior (Gibbs row): $q_1^* =$ N$(B/A, 1/A)$ with μ_2 replaced by $\mathbb{E}\mu_2$.

Derivations to copy: N$(\mu, 1/\tau)$ data, priors as Gibbs box.
Gibbs μ: $p(\mu|\tau,x) \propto \exp\{-\frac{\tau}{2}\sum(x_i - \mu)^2 - \frac{\kappa}{2}(\mu - m)^2\} \propto \exp\{-\frac{1}{2}(n\tau + \kappa)\mu^2 + (\tau n\bar x + \kappa m)\mu\}$: $A = n\tau + \kappa$, $B = \tau n\bar x + \kappa m$.
CAVI τ: $\log q_\tau^* = \mathbb{E}_\mu[\log p(x|\mu,\tau) + \log p(\mu|\tau) + \log p(\tau)] + c = (\frac{n}{2} + \frac{1}{2} + a_0 - 1)\log\tau - \tau[b_0 + \frac{1}{2}\sum\mathbb{E}(x_i - \mu)^2 + \frac{\lambda_0}{2}\mathbb{E}(\mu - \mu_0)^2] + c$: power of $\log\tau$ is $a^* - 1$, coefficient of $-\tau$ is b^*.
MH, $x \sim$ N$(\mu, 1)$, $\mu \sim$ N$(0,1)$, **random walk:** $\log\frac{\pi(\mu')}{\pi(\mu)} = -\frac{1}{2}[(x - \mu')^2 + \mu'^2] + \frac{1}{2}[(x - \mu)^2 + \mu^2]$; $A = \min(1, e^{\text{this}})$.

MCMC vs VI: MCMC simulates from the density, is heavier, asymptotically exact. VI approximates the density, is faster, suits large data and quick model exploration, has no guarantee, generally underestimates variance.

Maid Mast Codes PT4: MCMC
- `prob<-dtarget(prop)/dtarget(cur); if(runif(1)<prob) accept`: symmetric MH. `is.nan` guard.
- `proposal<-rnorm(2,chain[i,],sd)`: random walk.
- `acceptance<-1-mean(duplicated(chain[-(1:burnIn),]))`: after burn-in. `chain[-(1:burnIn),]` drops burn-in.
- `la1<-rgamma(1,a1+cfreq[M],b1+M)`: Gamma(shape, rate); `cfreq<-cumsum(freq)` gives $\sum_1^m y_i$.
- `M<-sample(1:n,1,prob=(la1/la2)^cfreq*exp((la2-la1)*(1:n)))`: discrete conditional.
- `rnorm(1,mu1+(x2-mu2)*s12/s2,sqrt(s1-s12^2/s2))`: Gibbs, sd not var.
- `cumsum(x)/1:n`: running mean, should flatten. `mean(th>0)`: posterior probability. `s[seq(1,n,by=k),]`: thin.

Old-paper extras: **Bayes estimator:** minimise posterior expected loss; squared error gives posterior mean. **Bayesian GLM:** vague priors $\beta_j \sim$ N$(0, 10^4)$; overdispersion: add $\varepsilon_i \sim$ N$(0, 1/\tau)$ to η_i. **Log-linear:** Poisson on table counts; `A*B` = interaction; residual df = cells − parameters; `A+B+C` = mutual independence.

Exam checks:
- Define P_{ik}; state which θ^0.
- State H_0, statistic, df, decision, sentence in context.
- CI on η first, transform last.
- $\hat p$, $\hat\mu$ on response scale.
- p includes intercept.
- $\hat\phi$: big model for F, own model for SE.
- Rate vs scale; variance vs sd in N.
- Name distribution and every parameter.